

Presence Factors Scale (PFS): Application Guide and Complete Instrument

August 25, 2026

1 Introduction

The **Presence Factor Scale (PFS)** is a low-cost and highly applicable assessment method for measuring the sense of presence induced by a Virtual Environment (VE) or Virtual Reality (VR) experience. The PFS reduces the need for extensive and expensive user testing, serving primarily as an assessment tool and a set of design guidelines. The instrument consists of **23 factors**, identified through an extensive systematic literature review, organized into three main categories: **Engagement, Interactive Fidelity, and Sensory Fidelity**.

2 Scoring Methodology and Calculation

The **PFS Score** is an average value that quantifies the potential degree of presence of a VE. The final score is calculated as the **arithmetic mean** of the scores assigned to all 23 factors across the three main categories.

2.1 Score Assignment

For each of the 23 factors, the evaluator must select the **level** that best describes the characteristic implemented in the VE being evaluated. Each level has a predetermined score on a **1.0-7.0 scale** (where 7.0 represents the maximum contribution to presence).

2.2 Formula for PFS Score Calculation

The mathematical formula for calculating the PFS Score is presented as follows:

$$\text{PFS Score} = \frac{\sum_{i=1}^N S_i}{N} \quad (1)$$

Where:

- PFS Score: The final presence score of the Virtual Environment (result between 1.0 and 7.0).
- S_i : The score assigned to the i -th factor evaluated (value between 1.0 and 7.0).
- N : The total number of factors evaluated (for the PFS, $N = 23$).
- $\sum_{i=1}^N S_i$: Represents the sum of the scores assigned to all 23 factors.

2.3 Interpretation

A **higher PFS Score** indicates that the VE's design and technical characteristics offer a **greater potential to induce the sense of presence** in the user.

3 Complete Instrument: Factors, Levels, and Scores

Tables 1, 2, and 3 present the complete list of the 23 factors, their levels, and the corresponding scores (value between 1.0 and 7.0). To apply the PFS, select, for each factor, the level that best suits the Virtual Environment and record the corresponding score. A version of the instrument, with a description of each factor and its levels, is presented in Section 4.

Table 1: Factors of the Engagement Category

#	Factor	Levels (score)	Score
1	Inclusiveness	None (2.0), Partially (5.0), Totally (7.0)	2.0 - 7.0
2	Narrative	Incomplete (2.5), Complete (5.0), Engaging (6.5)	2.5 - 6.5
3	Attention	Alternating (2.5), Selective (5.0), Concentrated (6.0), Sustained (7.0)	2.5 - 7.0
4	Emotion	None (1.0), Weak (3.0), Medium (5.0), Strong (7.0)	1.0 - 7.0
5	Task	None (1.0), Simple (3.0), Medium (5.0), Complex (7.0)	1.0 - 7.0
6	Copresence	None (1.0), Partial (4.0), Total (7.0)	1.0 - 7.0
7	Social Presence	None (1.0), Low (3.0), Moderate (5.0), High (7.0)	1.0 - 7.0

Table 2: Factors of the Interactive Fidelity Category

#	Factor	Levels (Score)	Score
8	Manipulation	None (1.0), Low Intuition (4.5), Intuitive (5.5), Very Intuitive (7.0)	1.0-7.0
9	Travel	None (1.0), Low Intuition (2.0), Intuitive (5.0), Very Intuitive (7.0)	1.0-7.0

Table 3: Factors of the Sensorial Fidelity Category

#	Factor	Levels (Score)	Score
10	Invasiveness	High (1.5), Moderate (3.5), Low (5.5), None (7.0)	1.5-7.0
11	Proprioception	Absent (1.0), Little (2.5), Medium (4.5), Total (6.5)	1.0-6.5
12	Body Ownership	Absent (1.5), Partial (4.5), Total (7.0)	1.5-7.0
13	Latency	Above 20ms (1.5), 10-20ms (4.5), 00-10ms (7.0)	1.5-7.0
14	Transition	Instantaneous (4.0), Gradual (7.0)	4.0-7.0
15	Synchronization	Asynchronous (3.0), Synchronous (6.5)	3.0-6.5
16	Plausibility	Implausible (3.0), Plausible (6.5)	3.0-6.5
17	Visual Realism	None (1.0), Static (4.0), Dynamic (7.0)	1.0-7.0
18	Field of View	Up to 100°(3.0), Up to 130°(6.0), Up to 180°(7.0)	3.0-7.0
19	Resolution	LDPI (1.0), MDPI (2.0), TVDPI (3.0), HDPI (4.0), XHDPI (5.0), XXHDPI (6.0), XXXHDPI (7.0)	1.0-7.0
20	Frame Rate	Up to 60 FPS (2.5), 60-90 FPS (5.5), Greater than 90 FPS (7.0)	2.5-7.0
21	Audio	Mono (2.5), Stereo (5.0), Surround (6.0), 3D (7.0)	2.5-7.0
22	Haptics	Absent (1.0), Simple (4.5), Complex (6.5)	1.0-6.5
23	Olfaction	Absent (3.0), Present (6.5)	3.0-6.5

4 Description of Factors and their Levels

Tables 4, 5, and 6 present a description of the factors and their respective levels.

Table 4: Engagement Factors and Levels Description

#	Factor	Levels	Score
1	Inclusiveness - the degree to which physical reality is excluded.	None - interference from the physical environment occurs throughout the entire virtual experience.	2.0
		Partially - interference from the physical environment occurs at some moments.	5.0
		Totally - there is no interference from the physical environment.	7.0
2	Narrative - the story designed to be experienced virtually by the user.	Incomplete - it does not present all elements of the narrative (fact, time, place, characters, cause, mode, and consequence).	2.5
		Complete - presents all elements of the narrative (fact, time, place, characters, cause, mode, and consequence).	5.0
		Engaging - it has all the elements and includes at least one challenge, suspense, or surprising fact.	6.5
3	Attention - the degree to which the virtual environment retains the user's attention.	Alternating - allows the user to switch attention to activities outside the experience.	2.5
		Selective - allows the user to switch attention, but, due to its attractiveness, maintains focus.	5.0
		Concentrated - requires total and exclusive concentration from the user.	6.0
		Sustained - requires the user's attention for an extended period without distractions.	7.0
4	Emotion - the emotions (Anger, Sadness, Happiness, Disgust, Surprise, Fear) experienced by the user in the virtual environment.	None - the virtual environment does not cause emotions in the user.	1.0
		Week - contemplates situations that cause week emotion in the user.	3.0
		Medium - contemplates situations that cause medium emotion in the user.	5.0
		Strong - contemplates situations that cause strong emotion in the user.	7.0
5	Task - the act of doing something in the virtual environment.	None - the virtual environment requires no tasks beyond visualization.	1.0
		Simple - the virtual environment requires the execution of simple tasks.	3.0
		Medium - the virtual environment requires the execution of medium complexity tasks.	5.0
		Complex - requires the execution of complex tasks.	7.0
6	Copresence - the feeling of being with another in a virtual environment.	None - there are no other actors in the virtual environment.	1.0
		Partial - there are other actors at some points in the experience.	4.0
		Total - there are other actors throughout the entire experience.	7.0
7	Social Presence - the act of interacting with another in a virtual environment.	None - there is no interaction with other actors.	1.0
		Low - there is little interaction with other actors.	3.0
		Moderate - there is medium interaction with other actors.	5.0
		High - there is a lot of interaction with other actors.	7.0

Table 5: Interactive Fidelity Factors and Levels Description

#	Factor	Levels	Score
8	Manipulation - the ability of the virtual environment to be modified by user actions.	None - does not allow interaction.	1.0
		Low Intuition - enables non-intuitive interaction.	4.5
		Intuitive - enables intuitive interaction.	5.5
		Very Intuitive - enables very intuitive interaction.	7.0
9	Travel - the way to navigate in the virtual environment.	None - the virtual environment does not allow navigation.	1.0
		Low Intuition - the way to navigate in the VE is not very intuitive (e.g. teleportation).	2.0
		Intuitive - The way to navigate in the VE is intuitive (e.g. joystick, fly, walk in place).	5.0
		Very Intuitive - the way to navigate in the VE is very intuitive (e.g. vehicle, real walk).	7.0

Table 6: Sensorial Fidelity Factors and Levels Description

#	Factor	Levels	Score
10	Invasiveness - number of external devices required.	High - more than four devices arranged over the user.	1.5
		Moderate - three or four devices arranged over the user.	3.5
		Low - up to two devices arranged over the user.	5.5
		None - no devices against the user.	7.0
11	Proprioception - sense of self-movement and body position.	Absent - proprioceptive feedback not provided.	1.0
		Little - proprioceptive feedback poorly provided.	2.5
		Medium - average proprioceptive feedback.	4.5
		Total - proprioceptive feedback very provided.	6.5
12	Body Ownership - sense of body ownership in the virtual environment.	Absent - does not provide the sense of body ownership.	1.5
		Partial - provides partial sense of body ownership.	4.5
		Total - provides total sense of body ownership.	7.0
13	Latency - delay between user action and the reaction of the virtual environment.	Above 20ms - greater than 20 ms.	1.5
		10-20ms - between 10 and 20 ms.	4.5
		00-10ms - up to 10 ms.	7.0
14	Transition - change of state between the real and virtual world.	Instantaneous - abrupt transition.	4.0
		Gradual - gradual transition.	7.0
15	Synchronization - coordination between user actions and all virtual environment outputs.	Asynchronous - it lacks adequate visual, sound, and inter-action synchronization.	3.0
		Synchronous - it shows synchronization between all stimuli.	6.5
16	Plausibility - the extent to which virtual environment is admissible.	Implausible - unacceptable, inadmissible.	3.0
		Plausible - acceptable, admissible.	6.5
17	Visual Realism - to what extent does the virtual environment looks visually real.	None - does not provide geometric and lighting realism to objects.	1.0
		Static - provides realism to objects when stationary.	4.0
		Dynamic - provides realism to objects even when in motion or altered.	7.0
18	Field of View - the extent of the observable virtual world at any given moment.	Up to 100° - field of view up to 100°.	3.0
		Up to 130° - field of view up to 130°.	6.0
		Up to 180° - field of view up to 180°.	7.0
19	Resolution - number of individual dots that can be placed in a 1-inch line.	LDPI - low density, 120 dots per inch.	1.0
		MDPI - medium density, 120–160 dots per inch.	2.0
		TVDPI - medium high density, 160–213 dots per inch.	3.0
		HDPI - high density, 213–240 dots per inch.	4.0
		XHDPI - extra high density, 240–320 dots per inch.	5.0
		XXHDPI - extra extra high density, 320–480 dots per inch.	6.0
20	Frame Rate - number of unique images shown per second.	Up to 60 FPS - up to 60 frames per second.	2.5
		60-90 FPS - between 60 and 90 frames per second.	5.5
		Greater than 90 FPS - over 90 frames per second.	7.0
21	Audio - the way sound travels through the user’s head.	Mono - sound captured and reproduced by a single channel.	2.5
		Stereo - sound captured and reproduced by two channels (right and left) synchronized in time.	5.0
		Surround - multichannel sound system, 5 or 7 mono channels at the same time.	6.0
		3D - surround sound emulation using six to eight channels.	7.0
22	Haptics - sense of touch or be touched by something in the virtual environment.	Absent - does not provide tactile or force feedback.	1.0
		Simple - simple motor vibration.	4.5
		Complex - it provides force (kinesthetic) feedback and/or detailed tactile (cutaneous) stimulation.	6.5
23	Olfaction - the sensation of perceiving the smells.	Absent - does not provide olfactory feedback.	3.0
		Present - provides olfactory feedback.	6.5